

Capture Data Logger supports different types of channels like analogue input/output, digital input/output, thermocouple input, strain input, vibration input and CAN input to name a few. It also supports configuration and viewing of calculated channels. The channel name, units and scale are user-configurable. Different scaling types like linear, polynomial, table and map ranges are supported by this software.

Filters such as band pass, band stop, low pass and high pass can be selected by the user as per the requirement. Other filters such as Butterworth, Chebyshev, Bessel and Elliptic are also user-selectable.

Capture VDAS supports different terminal configurations like reference single-ended or differential, and also supports different display types like graph, numeric, gauge, tank, slider, boolean and custom. It is capable of creating threshold triggers, time triggers and user-configurable triggers. It performs different types of analysis for vibration and strain. It can acquire and log channel data at different sampling rates. Offline data replay and trending of the data in graph and table is available.

Apart from data logging to TDMS, ASCII and binary files being available, it is possible to replay and analyse the logged data. It is also possible to transfer the data logged in the data logger to the laptop connected to it with the simple click of a button. It supports logging to multiple files at once, each with its own configuration and independent triggers. It is also possible to perform online analysis such as fast

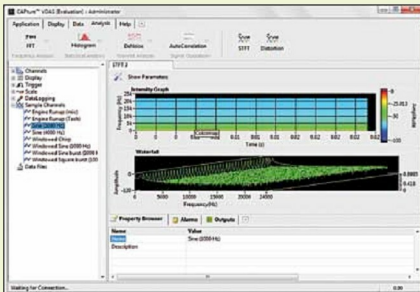


Fig. 3: STFT Analysis

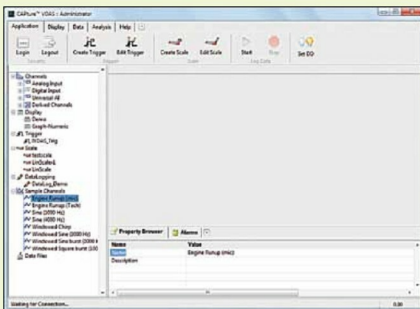


Fig. 4: Work Space

Fourier transform (FFT) and power spectrum on the acquired data.

Features in detail

Channel configuration. The software has the provision to configure the channels that are listed for ease of access. Any channel can be configured directly from the channel list or from the respective parameter configuration window.

Display. Any channel from the channels list can be dragged and dropped onto the displays. The user can launch different displays, position them, save the display configuration and load the display configuration from a file.

Trigger. The software has the pro-

vision to configure the triggers. The user can create a trigger, edit a trigger and delete a trigger. In a threshold trigger, the user can select the channel name, condition and enter the threshold, whereas in the case of time trigger, the user can simply enter the time in seconds.

Scaling. The software allows the user to configure different types of scaling such as linear scaling, polynomial scaling, table scaling and map ranges. The user can create a scaling, edit a configured scaling and delete a scaling configuration.

Data logging. Data logging configuration consists of the trigger information, display information and the channels to be logged. The user can configure a data logging configuration by selecting a trigger from a list of triggers configured, the channels to be logged from the available list of channels or a display type from the

different displays configured.

Offline. Offline utilities include file transfer utility, replay data and view data. The software includes different types of analysis for vibration and strain data. The offline analysis module can be customised as per the customer requirement.

Applications

CAPTure Data Loggers can be used in a wide variety of applications like in-vehicle data logging, in-flight data logging, general research, remote data logging, environmental monitoring and power monitoring. ●

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